

A267 – The Stipulation $\alpha = 1$: Charge Redefinition and Conversion Examples

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.1 Purpose

[STIPULATION] In FFGFT α follows from the geometry (A130). This document treats the complementary question: how does one set $\alpha = 1$ *as a computational tool* in a way that is fully reversible and makes the coupling structure visible? The answer is the Heaviside–Lorentz stipulation $e^2 = 4\pi\alpha$ combined with $\alpha = 1$ – not a unit choice but a *stipulation* with preserved trace.

.2 Stipulation: Heaviside–Lorentz with $\alpha = 1$

[STIPULATION] In natural units ($\hbar = c = 1$) and the Heaviside–Lorentz system, the *defining equation* is

$$e^2 = 4\pi\alpha.$$

The fine-structure constant is set to $\alpha = 1$. This is not a *norm* (unit choice) – a dimensionless number is invariant under rescaling of base units – but a *stipulation*: the coupling is adopted as a computational unit.

[B] The charge carries the 4π ; what is set to 1 is α , not e . From $e^2 = 4\pi\alpha$ with $\alpha = 1$ follows the value $e = \sqrt{4\pi} \approx 3.545$ – not $e = 1$. The relation $e^2 = 4\pi\alpha$ remains as a defining equation and cannot be dissolved by the stipulation.

.3 Back-Computation as Substitution

[B] Because $e^2 = 4\pi\alpha$ remains as a relation, the coupling power is automatically carried along. Back-computation is pure substitution

$$\alpha \longrightarrow \frac{1}{137.036},$$

applied to the α -order of the process:

- per vertex a factor $e = \sqrt{4\pi\alpha}$; compared to the $\alpha=1$ value $\sqrt{4\pi}$, that is a $\sqrt{\alpha}$ back,
- amplitude with V vertices $\propto (4\pi\alpha)^{V/2} \Rightarrow \times\alpha^{V/2}$,
- rate/cross section $\propto (4\pi\alpha)^V \Rightarrow \times\alpha^V$,
- compact: process of rate order $\alpha^n \Rightarrow \times\alpha^n$.

[B] The 4π is not double-counted: it is already in e^2 and in the phase-space normalisation. Only α^n is reset, not $(4\pi\alpha)^n$.

.4 Conversion Examples

[K] All quantities in $\hbar = c = 1$, Heaviside–Lorentz. Back-factor = the α -order of the expression.

Quantity	Physical form	With $\alpha=1$	Back-factor
Coulomb potential	$V = \alpha/r$	$1/r$	$\times\alpha$
Class. electron radius	$r_e = \alpha/m_e$	$1/m_e$	$\times\alpha$
Compton wavelength	$\lambda_C = 1/m_e$	$1/m_e$	$\times 1$
Bohr radius	$a_0 = 1/(\alpha m_e)$	$1/m_e$	$\times\alpha^{-1}$
Rydberg / binding	$E = \frac{1}{2}\alpha^2 m_e$	$\frac{1}{2}m_e$	$\times\alpha^2$
Thomson cross section	$\sigma_T = \frac{8\pi}{3}\alpha^2/m_e^2$	$\frac{8\pi}{3}/m_e^2$	$\times\alpha^2$

.5 The 137-Ladder: What $\alpha = 1$ Makes Visible

[B] The three atomic length scales stand in the ratio

$$r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}.$$

At $\alpha = 1$ they all collapse to $1/m_e$: they are not three independent scales but *one* scale with three α -powers. a_0 is exactly 137× larger than λ_C , and λ_C exactly 137× larger than r_e .

[B] With $\alpha = 1/137$ written out everywhere, this common origin remains hidden – three different awkward numbers appear and the α -power as ordering principle is not visible. The $\alpha = 1$ stipulation reveals it.

.6 Why This Stipulation – Comparison with Alternatives

[S] Versus Gaussian ($e^2 = \alpha$). Also works, but places the 4π in the field equations rather than in Coulomb’s law. Heaviside–Lorentz places the 4π where they geometrically belong (sphere surface, Coulomb) and leaves the Maxwell equations 4π -free. α is then the pure coupling, not a mixture of coupling and geometry.

[S] Versus $c = \hbar = 1$ unit choice. The setting $c = \hbar = 1$ destroys the powers – the trace is gone, back-computation runs only via dimensional analysis (A266). The defining equation $e^2 = 4\pi\alpha$, by contrast, *retains the trace*: the coupling power is carried along because the relation stands. Stipulation with preserved trace, not norm with erased trace – hence constructively reversible, not merely justifiably reconstructable.

.7 FFGFT Connection: Back-Computing α Means Inserting ξ

[B] α is in the FFGFT corpus not a free input but follows from the geometry (A130):

$$\alpha^{-1} = 137.036 \quad \text{from} \quad \xi = \frac{4}{30000}, \quad D_f = 3 - \xi, \quad E_0 = \sqrt{m_e m_\mu}.$$

The stipulation $\alpha = 1$ isolates precisely the point at which the ξ -geometry enters. Every reset α^n is a location at which $D_f = 3 - \xi$ becomes visible.

[B] Within the corpus, “back-computing α ” therefore does not mean “inserting $1/137$ ” but “inserting the ξ -structure” – and the $\alpha=1$ system makes countable at how many places and in which power that occurs.

.8 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$e^2 = 4\pi\alpha$	Heaviside–Lorentz defining equation
$e = \sqrt{4\pi} \approx 3.545$	Elementary charge at $\alpha = 1$
$r_e = \alpha/m_e$	Classical electron radius
$\lambda_C = 1/m_e$	Compton wavelength
$a_0 = 1/(\alpha m_e)$	Bohr radius
V	Number of Feynman vertices in the process
α^n	α -order / back-factor of the process

.9 Verification Script

- Numerical verification of all table rows with $\alpha = 1/137.036$ and $m_e = 0.511$ MeV; ratio $r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}$: `python/A_Serie_Skripte/a267_alpha_eins.py`

.10 Open Questions

The stipulation $\alpha = 1$ is fully worked out only for electromagnetic processes. For weak and strong coupling (α_W, α_s) an analogous table would be possible but has not been worked out.

The connection α^n as locus of $D_f = 3 - \xi$ is plausibly shown for simple QED processes but not for multi-loop or non-abelian processes.

.11 Supersedes

This document subsumes: Dok. 011 (individual formulas r_e , λ_C , a_0 , dimensional analysis $\alpha = r_e/\lambda_C$, SI numerical values of atomic scales), Dok. 015 (Heaviside–Lorentz line $e^2/(4\pi) = 1$ in the unit systematics). The coherent conversion table with explicit α^n back-factors and the 137-ladder $r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}$ as a structural statement are not present as a unit in the legacy corpus and are newly systematised here. Related: A130 (geometric derivation α from ξ), A266 (SI back-computation).

.12 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.